Saint Petersburg State

University Graduate School of Management

End-to-end Sample ML Project course

# Report 2: Data Preparation

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## Data Types Overview

| Feature | Dtype | Preprocessing Applied |
|---|---|---|
| year | int64 | Kept as-is, then StandardScaler |
| make | object | Hash Encoding (16 features) |
| model | object | Hash Encoding (16 features) |
| trim | object | Hash Encoding (16 features) |
| body | object | Lowercase, group rare, One-Hot Encoding |
| transmission | object | One-Hot Encoding |
| vin | object | DROPPED - unique identifier |
| state | object | One-Hot Encoding |
| condition | float64 | Imputed, then StandardScaler |
| odometer | float64 | Imputed, outlier removal, StandardScaler |
| color | object | One-Hot Encoding |
| interior | object | One-Hot Encoding |
| seller | object | Hash Encoding (16 features) |
| sellingprice | float64 | TARGET - kept as-is |
| saledate | object | DROPPED - text date format |

## Data Cleaning

- Removed rows where transmission contained 'Sedan' or 'sedan' - clear data entry errors
- Standardized make capitalization using .str.capitalize() - collapsed duplicates like 'ford', 'Ford', 'FORD'
- Lowercased body column and grouped rare body types (< 50 occurrences) into 'other'
- Removed top 1% of sellingprice values (above ~$44,500) - extreme outliers that skew regression
- Removed top 1% of odometer values - vehicles with 999,999 miles are likely data errors

Shape after cleaning: 48,514 rows x 13 columns (before encoding)

## Missing Values Imputation

Imputing strategies were applied rather than simply dropping or replacing with 'Unknown':

- condition: group-level median per make brand - premium brands (BMW, Ferrari) score higher than budget brands
- odometer: global median imputation
- transmission: mode imputation (automatic dominates ~97% of rows)

- make, model, trim, body, color, interior: filled with 'unknown' to preserve missingness signal
- sellingprice: rows dropped (only 4 missing)

## Feature Encoding

### Categorical Features - One-Hot Encoding

Applied to features with manageable unique values:

- body - after grouping rare types, 21 meaningful categories
- transmission - 2 categories (automatic, manual)
- state - 51 US states
- color - 16 exterior colors
- interior - 12 interior colors

### Categorical Features - Hash Encoding

Applied to high-cardinality features (hundreds of unique values):

- make - 64 unique brands -> 16 hash features
- model - 900+ models -> 16 hash features
- trim - 1200+ trims -> 16 hash features
- seller - 5000+ sellers -> 16 hash features

### Numerical Features

- year, condition, odometer - kept as-is, then StandardScaler applied
- sellingprice (target) - kept as-is, no scaling

Final dataset shape after encoding: 48,514 rows x 168 columns

## Feature Scaling

StandardScaler was applied to numerical features (year, condition, odometer):

- year: mean=2009.99 -> 0.00, values scaled to approximately [-6, +1]
- condition: mean=30.58 -> 0.00, values scaled to [-2.5, +1.5]
- odometer: mean=69,019 -> 0.00, right-skewed distribution preserved

Scaling centers all features to mean=0 and std=1, which is important for distance-based and linear models.

## Preprocessing Summary

| Step | Action | Result |
|---|---|---|
| **Data cleaning** | Fixed typos, grouped rare values, removed outliers | 48,514 rows |
| **Missing values** | Median/mode imputation + unknown fill | 0 missing values |
| **Drop columns** | Removed vin and saledate | -2 columns |
| **OHE** | body, transmission, state, color, interior | +102 columns |
| **Hash Encoding** | make, model, trim, seller | +64 columns |
| **StandardScaler** | year, condition, odometer | scaled |
| **Final dataset** | Saved to data/processed/data_prepared.csv | 48,514 x 168 |